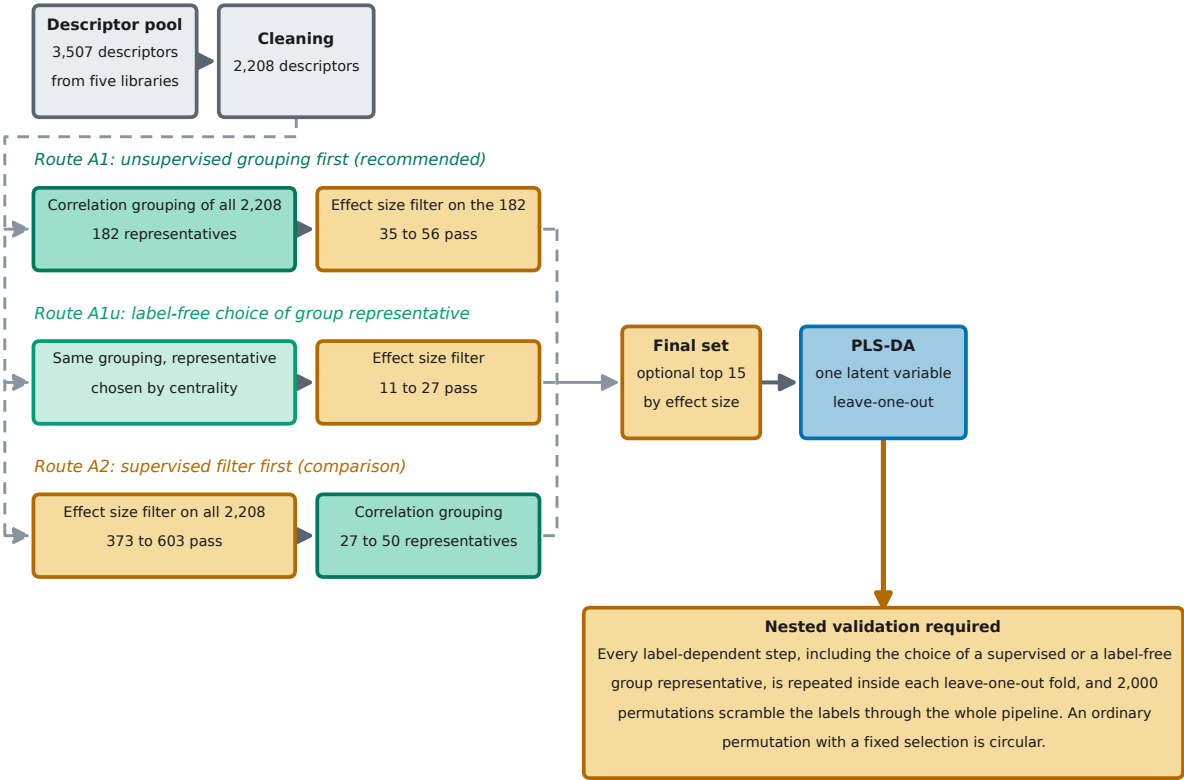
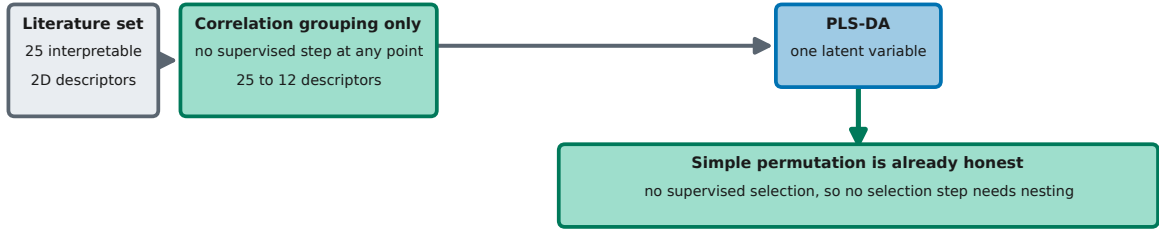


(a) Route A: large data-driven descriptor pool

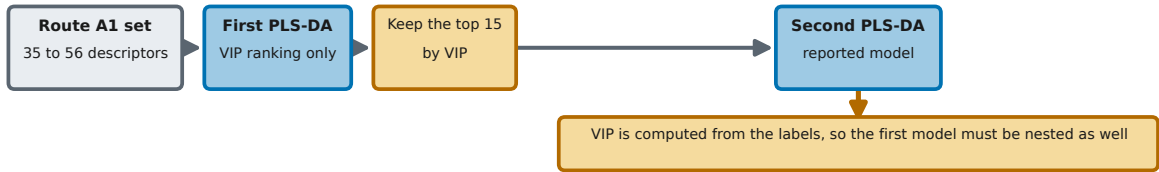


(b) Route B: small literature-derived descriptor set



(c) VIP-based selection

The first PLS-DA only ranks descriptors by VIP; the second, fitted on the top 15, is the model whose performance is reported.



- Supervised step: uses the activity labels, source of selection leakage
- Unsupervised step: uses descriptor values only
- Route A1u: label-free choice of representative
- PLS-DA model
- Data step without descriptor selection